Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/27/2006
PI Name:	Ho, Tin-Lun (Jason) Ph.D.		
Project Title:	Quantum gases in novel environments: Optical lattices and rapidly rotating potentials.		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Columbus	**State:**	OH
Zip Code:	43210	**Congressional District:**	15
Comments:	P.I. Web page will be set up in the next few months		
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	11/01/2001	**End Date:**	01/31/2006
No. of Post Docs:	3	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG 3 2916		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		

The objective of this task is to study the properties of Bose-Einstein condensates and highly degenerate Fermi gases in novel environments such as optical lattices and rapidly rotating potentials. These environments have been realized recently, and many experiments are being performed on these systems today. Because of the close connection between these systems and electron systems in solids, the study of these systems will help to address many fundamental issues in condensed matter physics. Furthermore, experimental discoveries in both areas will help to cross-fertilize technological advances in both fields, especially in the processing of quantum information. Our goal is to study the properties of degenerate quantum in optical lattices and rapidly rotating potentials. These are analogs of some of the most intriguing systems in solid state physics, such as Hubbard models quantum spin, and quantum Hall systems. Our plan is to uncover the variety of new ground states in of cold atoms in these settings, to study their dynamics, and their quantum phase transitions. In particular, we shall focus issues that are fundamental to many different areas such as quantum optics,

Task Description:	condensed matter, and atomic physics, and nuclear physics. These include the effect of large scattering length, very fast rotating clusters, and various tunneling phenomena in optical lattices. Unlike the other task of mine, ``Gravitational effects on Bose-Einstein condensate of atomic gases", which focus on gravitational effects, this task focuses on the physics in optical lattices and rotating potentials. The issues addresses in this task (such as the nature of ground state, the collective phenomena, and the quantum phase transition between different phases) are not only important for understanding the properties of quantum gases, but also for understanding a number of key phenomena in condensed matter physics such as metal-insulator transition, quantum Hall effects with internal degrees of freedom. At the same time, the physics of rapidly rotating quantum gases is also related closely to that of rotating nuclei. The projects in this task are therefore not only relevant to the area of quantum gases, but also condensed matter and nuclear physics. Since many results emerged from this task are directly related to experiments in many laboratories in US and abroad, they have been subjects of invited talks in many recent international conferences.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)